

Highlights

Do Transportation Policies Affect Men and Women Equally? A Gender-Based Sensitivity Analysis of Urban Transport Mode Shifts

- Gender-neutral transport policies risk reinforcing mobility inequalities.
- Fare subsidies, security investments, and road safety campaigns must be designed and evaluated with gender disaggregation.
- Personal security improvements generate stronger modal shifts toward public transport among women than among men.
- Women are more sensitive than men to cost increases in private modes.
- Men respond more strongly to road safety improvements, particularly for motorcycles

Do Transportation Policies Affect Men and Women Equally? A Gender-Based Sensitivity Analysis of Urban Transport Mode Shifts^{*}

ARTICLE INFO

Keywords:

Gender Gap
Public Transportation
Discrete Choice Model
Elasticity
Mode Choice
Sensitivity Analysis
Mixed Logit Model

ABSTRACT

Urban transport policies are rarely evaluated through a gender lens, yet men and women respond differently to the same policy interventions. This paper presents a gender-disaggregated sensitivity analysis of urban transport mode choice, examining how changes in personal security perception, road safety perception, and travel cost-to-income affect the probability of choosing between cars, motorcycles, and public transport. Using survey data from Cali, Colombia — a motorcycle-dependent city representative of the mobility dynamics observed in cities across developing countries — we estimate a Mixed Logit model that allows for taste heterogeneity and realistic substitution patterns across modes. Our results reveal systematic gender asymmetries with direct policy relevance: personal security is the most powerful lever for modal shifts, with improvements in public transport security generating stronger substitution effects among women than among men. Women are also more sensitive to cost increases in private modes, reflecting tighter financial constraints rooted in structural inequalities. Conversely, men respond more strongly to road safety improvements, particularly for motorcycles. These findings demonstrate that gender-neutral transport policies are likely to reinforce rather than reduce mobility inequalities. The analytical framework proposed here offers a replicable reference for other territories seeking to embed a gender lens into transport policy evaluation.

1. Introduction

The challenges in urban mobility have prompted policymakers to search for strategies promoting sustainable transport modes, particularly public transportation. Policies such as fare incentives have been analyzed as mechanisms to encourage public transit use and discourage shifts toward private alternatives like cars and motorcycles (Sharaby and Shiftan, 2012; Myung-Jin, Ji-Eun and Hyun-Ju, 2018). Motorcycles are especially relevant in developing countries, where they offer affordable, convenient, and fast commuting options for working classes (Hagen, Pardo and Valente, 2016).

Evidence reveals significant gender differences in transport mode usage: men are more likely to use private vehicles, whereas women tend to rely on lower-cost alternatives, particularly public transportation (Dobbs, 2005; Vance and Iovanna, 2007; Ng and Acker, 2018; Basarić, Vujičić, Simić, Bogdanović and Saulić, 2016; Rosenbloom, 2006). However, gendered mobility patterns vary considerably across regional and cultural contexts. In many European cities, women prefer active modes and public transit, supported by safe infrastructure and progressive gender norms (Pirra, Kalakou, Carboni, Costa, Diana and Lynce, 2021). Urban policies in these regions prioritize pedestrian-friendly environments and integrated public transportation systems that facilitate women's participation in sustainable modes (e.g., bicycles) (Basbas, Papagiannakis, Baraklianos, Niki-foriadis and Campisi, 2023). In contrast, Latin American cities present a markedly different landscape. Transportation users face serious safety concerns, limited coverage, and unreliable service frequencies (Goel, 2023; Vanderschuren, Phayane and Gwynne-Evans, 2019; Hagen et al., 2016).

Combined with lower purchasing power for car ownership, this situation has driven increased reliance on motorcycles and informal transit options—which often serve as substitutes for formal systems and expose users to insecurity and harassment (Hagen et al., 2016). This context exacerbates vulnerability and restricts travel autonomy, particularly for women (Pirra et al., 2021).

Most literature on gender differences in urban mobility relies on qualitative (Rosenbloom and Burns, 1993; Hine and Mitchell, 2001) and descriptive approaches (Li, Guensler and Ogle, 2004; Dobbs, 2005; McGuckin, Zmund and Nakamoto, 2005; Kwan, 2005; Dobbs, 2007; Basarić et al., 2016; Elias, Newmark and Shiftan, 2008; Elias, Benjamin and Shiftan, 2015; Golob and Hensher, 2007; Taylor, Ralph and Smart, 2015; Zhang and Song, 2024; Adeel, Yeh and Zhang, 2016), examining variables such as daily trip frequency (Havet, Bayart and Bonnel, 2021), travel distance and duration, driver participation rates, household travel patterns (Craig and van Tienoven, 2019), commute stop frequency (Cao, Mokhtarian and Handy, 2008), and time allocation to work and care activities (Srinivasan, 2005; Fan, 2017). However, comparatively few studies explore how gender disparities in travel behavior influence decision-making processes underlying transport mode choice.

The limited research addressing this gap relies primarily on Discrete Choice Models (DCMs), especially the Multinomial Logit (MNL) model, valued for computational tractability and interpretability. Namgung and Akar (2014) demonstrated that attitudes toward public transit significantly influence transit choice, while Ng and Acker (2018) found that women travel shorter distances, prefer flexible modes like taxis, and rely more on public transportation than men. Yet these studies reveal a critical methodological gap: sensitivity analyses assessing how mode choice responds

^{*}

ORCID(s):

to policy changes are rare, and those conducted typically use the MNL model—which has well-documented limitations. Specifically, the MNL model cannot capture taste heterogeneity and imposes unrealistic substitution patterns through the Independence of Irrelevant Alternatives (IIA) assumption, where attribute changes in one alternative affect all others proportionally (Salazar-Serna, Barona, García, Cadavid and Franco, 2025). These limitations have prompted research on mode choice sensitivity to adopt more flexible models, including nested logit (S. and Chalumuri (Ch.), 2016), nested logit with latent classes (Wen, Wang and Fu, 2012), and mixed logit (Hensher and Greene, 2003). However, these methodologically superior approaches have not yet been applied to gender-differentiated mode choice analysis, representing an important gap in the literature. Nasrin and Chowdhury (2024) is a recent work on mobility that provides a gender-focused sensitivity analysis. Nevertheless it remains constrained by MNL model's limitations, further highlighting the need for more sophisticated approaches that simultaneously address gender differentiation and methodological rigor.

Since urban mobility is not gender-neutral and gender is a crucial determinant of travel behavior (Havet et al., 2021; Hanson, 2010), understanding gender-differentiated responses to transport attribute changes is essential for evidence-based policymaking. This study addresses this gap by analyzing gender-differentiated policy impacts through sensitivity analysis using a Mixed Logit (MXL) model. Using a Colombian city as a case study representative of developing-country transport dynamics, we evaluate three policies: (1) increased personal security perception; (2) increased road safety perception; and (3) travel cost changes. By centering gender in our analysis, we contribute to more policy-relevant sensitivity analyses that support equitable urban transport design aligned with women's mobility needs.

Cali, Colombia, provides an ideal case study for examining gender-differentiated mode choice in a developing-country context. As one of Latin America's most congested cities (TomTom, 2025), Cali exhibits characteristics typical of developing cities (Salazar-Serna, Cadavid, Franco and Carley, 2023): high accident rates (Agencia Nacional de Seguridad Vial, 2025), high motorcycle prevalence (Asociación Nacional de Empresarios de Colombia, 2024), and low public transit service quality perception (Alcaldía de Santiago de Cali, 2024). The city's formal public transport system has suffered from service deterioration, low coverage, and concerns over personal security, especially for women traveling during non-peak hours or in peripheral neighborhoods. This decline has led to sharp increases in motorcycle use, including among women users (Noguera, 2017). Where public transit no longer serves as the default or safest option for women, Cali exemplifies a context where gender-differentiated mode choices reveal actual disparities, making it ideal for testing sensitivity analyses in realistic developing-country conditions.

The remainder of this paper is organized as follows: Section 2 describes the discrete choice models and elasticity measures applied in the sensitivity analysis, along with the datasets and information gathered for the case study. Section 3 presents the results and discussion, and Section 4 provides conclusions and policy recommendations.

2. Materials and Methods

2.1. Methodology

2.1.1. Random Utility Model

Consider a transport mode choice problem with N decision-makers and J available alternatives. Within the Random Utility Models (RUM) framework, each decision-maker $n \in \{1, \dots, N\}$ derives a utility U_{nj} from choosing the alternative $j \in \{1, \dots, J\}$ and selects the transport mode that yields the highest utility, such that transport mode j is chosen if and only if $U_{nj} > U_{ni}$, for all $i \neq j$. Although utility is a latent construct, it is assumed to be decomposed into two components: $U_{nj} = V_{nj}(X_{nj}, Z_n) + \varepsilon_{nj}$, where $V_{nj}(X_{nj}, Z_n)$ represents the systematic (or representative) component of utility, which is a function of alternative-specific attributes X_{nj} and case-specific (or alternative-invariant) attributes Z_n , while the residual term ε_{nj} accounts for unobserved factors.

In this study, we set $J = 3$, representing three transport modes: public transport, car, and motorcycle—the predominant modes in most developing countries. The representative utility component $V_{nj}(X_{nj}, Z_n)$ is specified as follows. Alternative-specific variables, which vary across both individuals and alternatives, include the ratio of total travel cost (USD/trip) to daily labor income (USD/day) ($cost/income_{nj}$), perceived risk of traffic accidents ($safety_{nj}$), and perceived personal security ($security_{nj}$). Including the cost-to-income ratio captures the differentiated impact of travel expenses according to income levels, under the assumption that lower-income decision-makers are likely to assign greater importance to cost considerations. Case-specific variables, which vary only across decision makers but not alternatives, include gender ($women_n$), travel distance ($dist_n$), age (age_n), education level ($educ_n$), occupation status ($occup_n$), purpose of trip ($purpose_n$), number of stops ($stops_n$), and municipality of residence (mun_n).

Additionally, we incorporate case-specific variables that moderate the effect of alternative-specific attributes. These variables reflect how much importance respondents assign—measured on a five-point Likert scale—to travel cost ($cost_weight_n$), road safety ($safety_weight_n$), and personal security ($security_weight_n$). These factors interact with their corresponding variables to account for preference heterogeneity. For instance, individuals who report high concern for travel costs will experience greater disutility from expensive modes compared to those less concerned. Thus, the representative utility function is specified in a linear form as follows:

$$\begin{aligned}
V_{nj}(X_{nj}, Z_n) = & \alpha_j + \beta_1 \cdot \text{cost_weight}_n \cdot \left(\frac{\text{cost}_{nj}}{\text{income}_{nj}} \right) \\
& + \beta_2 \cdot \text{security_weight}_n \cdot \text{security}_{nj} \\
& + \beta_3 \cdot \text{safety_weight}_n \cdot \text{safety}_{nj} \\
& + \gamma_{1j} \cdot \text{women}_n + \gamma_{2j} \cdot \text{age}_n \\
& + \gamma_{3j} \cdot \text{dist}_n + \gamma_{4j} \cdot \text{educ}_n \\
& + \gamma_{5j} \cdot \text{stops}_n + \gamma_{6j} \cdot \text{occup}_n \\
& + \gamma_{7j} \cdot \text{purpose}_n + \gamma_{8j} \cdot \text{mun}_n.
\end{aligned} \quad (1)$$

Thus, the probability that decision-maker n chooses transport mode j , P_{nj} , is given by

$$P_{nj} = \Pr(U_{nj} > U_{ni}, \forall i \neq j) \quad (2)$$

$$= \Pr(\epsilon_{ni} - \epsilon_{nj} < V_{nj} - V_{ni}, \forall i \neq j). \quad (3)$$

This probability represents a cumulative distribution that expresses the probability that the random term $\epsilon_{ni} - \epsilon_{nj}$ falls below the representative component $V_{ni} - V_{nj}$. As shown by Train (2009), defining $f(\epsilon_n)$ as the joint density of the random vector $\epsilon_n = (\epsilon_{n1}, \dots, \epsilon_{nJ})$, this probability can be expressed as:

$$P_{nj} = \int_{\epsilon} \mathbb{1}(\epsilon_{ni} - \epsilon_{nj} < V_{nj} - V_{ni}, \forall i \neq j) f(\epsilon_n) d\epsilon_n. \quad (4)$$

2.1.2. Discrete Choice Models

Based on the expression (4), the MNL model, one of the most widely used discrete choice models in the literature for analyzing the determinants influencing urban transport mode choice, can be derived by assuming that ϵ_{nj} is independently and identically distributed following a Type-I-generalized extreme value distribution—or Gumble distribution—. Under this assumption, the choice probability is obtained by:

$$P_{nj} = \int \left(\prod_{i \neq j} \exp(-\exp(-(\epsilon_{nj} + V_{ni} - V_{nj}))) \right) \exp(-\epsilon_{nj}) \exp(-\exp(-\epsilon_{nj})) d\epsilon_{nj}. \quad (5)$$

As established by McFadden (1974), equation (5) reduces to the following closed-form expression of the choice probabilities, hereafter referred to as the logit probabilities L_{nj} :

$$P_{nj} = L_{nj} = \frac{\exp(V_{nj}(X_{nj}, Z_n))}{\sum_i \exp(V_{ni}(X_{ni}, Z_n))}. \quad (6)$$

Despite its widespread use due to computational simplicity and closed-form expression for choice probabilities (Domencich and McFadden, 1975), the MNL model exhibits

three main limitations: first, it does not account for random taste heterogeneity; second, its IIA assumption imposes restrictive substitution patterns; and third, it is unsuitable for panel data when observed factors are correlated over time for each decision-maker. To address these limitations, this study employs the MXL model. Unlike the standard logit model, in which the ratio P_{nj}/P_{ni} depends solely on alternatives j and i , the MXL formulation allows for more flexible substitution patterns by ensuring that this ratio is influenced by the entire set of available alternatives.

Under the random coefficients interpretation, the vector of parameters β varies across decision-makers according to a density function $f(\beta | \theta)$ which is parameterized by θ . Formally, the MXL model is expressed as a weighted average of the standard logit formula at different parameter values of β , where the weights are given by $f(\beta | \theta)$. Thus, given a decision-maker n , the probability of choosing transport mode j is given by:

$$P_{nj} = \int L_{nj} f(\beta | \theta) d\beta, \quad (7)$$

where L_{nj} denotes the logit probability expressed in (6), assuming that the representative utility V_{nj} follows a linear specification, as defined in (1). Equivalently, the MXL model expressed in (7) can be interpreted as a mixture of logit probabilities evaluated at different values of β with $f(\beta | \theta)$ serving as the mixing distribution (Train, 2009). Moreover, the MNL model emerges as a special case of the MXL model when the mixing distribution is degenerate at fixed parameter values.

In this study, the parameters associated with the alternative-specific variables are assumed to be random. The coefficients corresponding to the cost-to-income ratio, the safety relevance factor, and the security relevance factor are assumed to follow a zero-bounded triangular (ZBT) distribution, in line with prior research on travel cost analysis (León and Miguel, 2017; Kremer, Leino, Miguel and Zwane, 2011).

2.1.3. Sensitivity Analysis

Due to its more flexible substitution patterns, the MXL model is employed to assess the sensitivity of mode choice through direct and cross marginal effects and elasticities, quantifying how the probability of choosing a given transport mode responds to variations in its own attributes or those of competing alternatives. In particular, marginal effects capture the change in probability of choosing a specific alternative resulting from a unitary change in an attribute. For a decision-maker n and an alternative j , the direct Marginal Effect (ME) of a unitary change in the m -th attribute, denoted as $x_{nj}^{(m)}$, is defined as:

$$ME_{j, x_{nj}^{(m)}} = \frac{\partial P_{nj}}{\partial x_{nj}^{(m)}}, \quad (8)$$

which, following Zeigermann (2024) and as formulated in equation (7) of the MXL model, is equivalent to:

$$ME_{j,x_{nj}^{(m)}} = \int \beta^{(m)} L_{nj} [1 - L_{nj}] f(\beta | \theta) d\beta. \quad (9)$$

Similarly, for an alternative $i \neq j$, the cross-marginal effect is defined as:

$$ME_{j,x_{ni}^{(m)}} = \int \beta^{(m)} L_{nj} L_{ni} f(\beta | \theta) d\beta, \quad (10)$$

where $\beta^{(m)}$ is the m -th element of the vector β . Unlike marginal effects, elasticity is a unit-free measure that describes, *ceteris paribus*, the percentage change in the probability of choosing an alternative in response to a percentage change in an attribute of either the same or a different alternative. Subsequently, we present individual-level elasticities—both direct and cross elasticities—derived from an MXL, computed for each decision-maker and alternative.

For any decision-maker $n \in \{1, \dots, N\}$, the direct elasticity of an alternative j with respect to a marginal change in attribute m of alternative j , $x_{nj}^{(m)}$, is defined as:

$$E_{j,x_{nj}^{(m)}} = -\frac{x_{nj}^{(m)}}{P_{nj}} \frac{\partial P_{nj}}{\partial x_{nj}^{(m)}}. \quad (11)$$

Since the probability P_{nj} is defined according to the MXL model, it follows that:

$$E_{j,x_{nj}^{(m)}} = -\frac{x_{nj}^{(m)}}{P_{nj}} ME_{j,x_{nj}^{(m)}}, \quad (12)$$

Similarly, for any decision-maker n , the cross elasticity of an alternative $i \neq j$ w.r.t. a marginal change in attribute m of alternative j , $x_{nj}^{(m)}$, is defined as:

$$E_{j,x_{ni}^{(m)}} = -\frac{x_{ni}^{(m)}}{P_{nj}} ME_{j,x_{ni}^{(m)}}. \quad (13)$$

In a discrete choice problem with N decision-makers and J alternatives, expressions (12) and (13) yield NJ^2 elasticities, comprising NJ individual direct elasticities and $NJ(J-1)$ individual cross-elasticities. A straightforward approach to reducing the number of expressions to J^2 is to compute average elasticities across the sample.

2.2. Inputs for data driven analysis

2.2.1. Study Area

As noted earlier, this research uses Cali for data-driven analysis. Cali is Colombia's third-largest city, located in the southwest with an estimated population of 2,285,099 inhabitants (DANE, 2018). Recognized for its economic dynamism and role as a regional hub for commerce and mobility (Urrea Giraldo and Candelo Álvarez, 2017), Cali

maintains close economic, social, and transportation ties with surrounding municipalities—Palmira, Candelaria, Jamundí, and Yumbo—with populations of 359,888; 94,202; 181,942; and 108,895 inhabitants, respectively (DANE, 2018). Integrated transportation systems in the region facilitate connectivity and promote constant flows of people and goods. Approximately 800,000 vehicles circulate daily in Cali (Secretaría de Movilidad, 2024), with cars comprising 68% of the fleet. In contrast, motorcycles predominate in surrounding municipalities, representing over 62% of vehicles (Registro Único Nacional de Tránsito, 2025).

The predominance of motorcycles reflects diverse socioeconomic and gender implications. In Colombia, 90% of households with motorcycles belong to lower socioeconomic strata (Asociación Nacional de Empresarios de Colombia, 2024), demonstrating that increased motorcycle use responds to lower acquisition and maintenance costs, combined with the ability to navigate traffic congestion on limited road infrastructure. In this context, motorcycles have become essential for mobility and economic survival for millions of Colombians. However, this phenomenon carries significant consequences: elevated road accident rates and risk exposure for users, substantial CO2 emissions, and gender inequities. Notably, the masculinization of motorcycle use restricts women's autonomous mobility, as they rely more heavily on public transportation (Pérez, 2019). CEPAL has documented that Latin America and the Caribbean maintain high traffic mortality rates, presenting urgent challenges for safer and more equitable transportation systems (CEPAL, 2019).

These conditions create additional barriers to mobility for women and vulnerable groups, intensifying inequalities in access to safe and efficient transportation. Although motorcycles have expanded mobility in surrounding municipalities, their predominance reveals critical deficiencies in inter-municipal transit integration with Cali's public system. Compounding these challenges, Cali's public transit (BRT-MIO) has experienced significant user migration to private vehicles in recent years (Noguera, 2017), further pressuring decision-makers to sustain system viability while promoting equity and sustainability.

2.2.2. Data

Survey data During the second quarter of 2024, a structured self-administered survey was used to gather information on urban travel behavior as well as the sociodemographic and economic characteristics of 1,147 voluntary respondents within the study area, namely Cali, Colombia, and its surrounding municipalities: Palmira, Yumbo, Candelaria, and Jamundí. For analysis purposes, only respondents whose trip origin or destination fell within the study area were retained ($n = 1,016$). Further details on participant selection are provided in the flow diagram in Figure 1¹.

¹A link to an interactive dashboard summarizing the complete survey data, including all 1147 responses before any exclusions, is available at <https://public.tableau.com/views/Tablero->

Table 1

Costs Associated with Private Transportation Modes (Car and Motorcycle)

Mode	Concept	Description	Value (COP)	Value (COP/km)
Car	Capital	Purchase ^a	45,771,111	305.14
		Useful life (years of use) ^b	10	
		Annual mileage (km/year) ^c	15,000	
	Taxes, insurance, and maintenance	Taxes ^d	686,566.665	45.77
		Compulsory insurance (SOAT) ^e	501,700	33.45
		Comprehensive insurance ^f	2,200,000	146.70
		Maintenance (estimated) ^g	8,238,799	549.25
	Fuel	Gasoline ^h	14,973 (COP/gallon)	467.91
Total				1,548.22
Motorcycle	Capital	Purchase ^a	5,630,750	62.56
		Useful life (years of use) ^b	6	
		Annual mileage (km/year) ^c	15,000	
	Taxes, insurance, and maintenance	Taxes ^d	84,461.25	5.63
		Compulsory insurance (SOAT) ^e	278,200	18.55
		Comprehensive insurance ^f	575,668.68	38.38
		Maintenance (estimated) ^g	1,520,302.50	101.35
	Fuel	Gasoline ^h	14,973 (COP/gallon)	136.00
Total				362.47

^a Source: Market prices were obtained from the database of new and used vehicle prices (national and imported) from Revista Motor (2023a), a leading automotive industry source in Colombia. Values correspond to a weighted average between new and used vehicles.

^b Source: The value was estimated based on the total usage time (in years) of the vehicle and its useful life (in years). A usage period of four to seven years is considered for vehicle replacement (La República, 2018; El Tiempo, 2022; Turbo, 2023), and a useful life of 18 to 20 years (El Espectador, 2024b; El Tiempo, 2024a)).

^c Source: A single value is assumed for private vehicles: (Caracol Radio, 2023; El Tiempo, 2024b; El Espectador, 2024a).

^d Calculated as 1.5% of the average purchase cost (El Ministerio de Hacienda y Crédito Público, 2023).

^e Source: Reference values for cars and motorcycles in 2023 (Revista Motor, 2023b). For cars, the reference value corresponds to a family car under 10 years old with an engine size of 1,500 to 2,500 CC; for motorcycles, the reference value corresponds to a motorcycle with an engine size of 100 to 200 CC.

^f Source: Reference value reported in El País (2023).

^g Source: Following a recent study, maintenance costs are assumed to be a proportion ($\gamma = 0.18$) of the capital cost (see Cuéllar-Alvarez, Clappier, Rojas-Roa, Thunis, Mangones and Belalcázar-Cerón (2024)).

^h Source: Reference price by city – effective November 4, 2023, reported by La Comisión de Regulación de Energía y Gas - CREG (2023). Regular gasoline: 14,973 COP/gallon. The calculation assumes an average car fuel efficiency of 32 km/gallon.

ⁱ Source: Market prices were obtained from the database of new motorcycle prices from Revista Motor (2023a), supplemented with used motorcycle prices from platforms such as Mercado Libre (2023); Tu Carro (2023); Carro Ya (2023). Values correspond to a weighted average between new and used vehicles.

^j Source: The value was estimated based on the total usage time (in years) of the motorcycle and its useful life (in years). A usage period of five to eight years is considered for vehicle replacement (La República, 2024), and a useful life of 10 to 15 years (El Espectador, 2024b).

^k Calculated as 1.5% of the average purchase cost (El Ministerio de Hacienda y Crédito Público, 2023).

^l Source: Value adjusted to 2023, based on information from SURA (2024).

^m Source: Following a recent study, maintenance costs are assumed to be a proportion ($\gamma = 0.27$) of the capital cost (Cuéllar-Alvarez et al., 2024).

ⁿ Source: Reference price by city – effective November 4, 2023, reported by La Comisión de Regulación de Energía y Gas - CREG (2023). Regular gasoline: 14,973 COP/gallon. The calculation assumes an average motorcycle fuel efficiency of 110 km/gallon.

Estimated travel costs Since travel costs were not directly reported in the survey, we estimated them for each individual based on their most frequent trip, using the reported origin and destination. Distances and travel times were calculated with Google Maps, assuming the use of a car, motorcycle, or public transport. When the origin and destination fell within different polygons (e.g., municipalities), distances were computed from centroid to centroid; otherwise, we used a centroid-to-boundary estimate.

The total travel cost per kilometer is computed as the sum of the use cost and the opportunity cost of travel time. For private transport modes (cars and motorcycles), the use cost per kilometer comprises expenses for fuel, maintenance, insurance, taxes, and capital depreciation. These estimates rely on market prices and usage parameters, with specific values and assumptions detailed in Table 1. For public transport, we applied fare structures relevant to intra- and inter-municipal trips. In both cases —private and public— we also incorporated the opportunity cost of time, calculated as the product of total travel time (including in-vehicle time and, for public transport, walking and waiting time).

DinamicasDeMovilidad_17380903114930/Dashboard?:language=es-ES&:
embed=y&:sid=&:redirect=auth&:embed_code_version=3&:loadOrderID=0&:
display_count=y&:origin=viz_share_link

Table 2

Sample and population composition comparison by gender.

Variable	Categories	Sample percentage	Population percentage
Gender	Women	53.94%	53.14%
	Men	46.06%	46.86%

3. Results and Discussion

This section presents the principal findings of the descriptive analysis for the applied MXL model. As we mentioned earlier, after the data cleaning step, we obtained information from 1,016 survey respondents (see also Figure 1). Table 2 compares the sample composition versus Census Data Projections (CDP) by gender in the urban area analyzed. The CDP were obtained from DANE (2018). Note that our sample is representative of the population by gender, which is our main comparative feature. It is important to mention that gender was considered based on participants' self-identification.

3.1. Descriptive Findings

3.1.1. Gender disparities in socioeconomic status and labor market participation

From Table 3, more than half of the respondents have attained education beyond the secondary level (50.39%), with a significantly higher proportion among women compared to men (54.01% vs. 46.15%). However, this higher educational attainment does not translate into improved socioeconomic outcomes for women. The sample is predominantly concentrated in low and middle socioeconomic status (SES) groups, with 45.28% in the low SES group and 42.22% in the middle SES group. Notably, women are disproportionately represented in the low SES group compared to men (49.82% vs. 39.96%), while men are overrepresented in the high SES group (16.24% vs. 9.31%). Occupational status reveals a substantial gender gap in labor market participation, with men showing higher rates of employment or self-employment compared to women (58.12% vs. 46.90%). Domestic work was reported exclusively by women (11.86%, $n = 65$), highlighting the gendered distribution of unpaid care responsibilities and its implications for the attachment to the labor market.

These patterns are consistent with previous research showing that women's higher educational attainment does not necessarily translate into improved labor market outcomes due to structural barriers and the unequal distribution of unpaid care work (Craig and van Tienoven, 2019; Havet et al., 2021). This imbalance constrains women's income and time availability, which in turn affects their ability to access mobility resources and shapes subsequent transport decisions (Adeel, Yeh and Zhang, 2017). Thus, gender inequalities in socioeconomic conditions are closely intertwined with disparities in mobility.

Table 3

Summary Statistics: Educational level and socioeconomic status by gender

Variable	Categories	Total	Women	Men
Educational Attainment	Primary	101 (9.94%)	55 (10.04%)	46 (9.83%)
	Secondary	403 (39.67%)	197 (35.95%)	206 (44.02%)
	Tertiary	512 (50.39%)	296 (54.01%)	216 (46.15%)
Socioeconomic Status	High	127 (12.5%)	51 (9.31%)	76 (16.24%)
	Low	460 (45.28%)	273 (49.82%)	187 (39.96%)
	Middle	429 (42.22%)	224 (40.88%)	205 (43.8%)
Occupation	Homemaker	65 (6.4%)	65 (11.86%)	—
	Unemployed or inactive	63 (6.2%)	27 (4.93%)	36 (7.69%)
	Employed or self-employed	529 (52.07%)	257 (46.9%)	272 (58.12%)
	Student	359 (35.33%)	199 (36.31%)	160 (34.19%)

Table 4

Transport modes in the sample by gender

Transport Modes	Gender		Total
	Women	Men	
Car	13.44%	16.34%	29.78%
Motorcycle	22.28%	22.88%	45.16%
Public Transit	16.71%	8.35%	25.06%
Total	52.42%	47.58%	100.00%

3.1.2. Transportation mode use and gendered access to mobility resources

The most common urban transport modes reported in the sample are private vehicles, specifically motorcycles (45.16%) and cars (29.78%), followed by public transit (25.06%) (see Table 4).

Significant gender differences emerge in modal choice patterns. Private vehicles are used more frequently by men than women—cars (54.9% vs. 45.1%)—while women rely more heavily on public transit (66.7% women vs. 33.3% men) (see Figure 3). Motorcycles are the most commonly used mode among both groups, though still more prevalent among men. Among women, 42.5% use motorcycles and 31.9% rely on public transit, whereas among men, 48.1% use motorcycles, 34.4% use cars, and only 17.6% use public transit (see Figure 2).

Furthermore, we found that a majority of respondents hold a driver's license (51.87%), but with a wide gender gap: 65.17% of men versus 40.51% of women are licensed drivers. These disparities reflect underlying inequalities in access to mobility resources. Rather than reflecting purely individual preferences, these patterns are consistent with evidence suggesting that women's mobility is constrained by limited access to transport assets and reduced control over household vehicles (Dobbs, 2005; Vance and Iovanna, 2007; Scheiner and Holz-Rau, 2012). Consequently, women's greater reliance on public transport is better understood as a reflection of constrained access rather than a simple modal preference.

3.1.3. Attitudes toward urban transport mode attributes

The attitudes of the respondents towards the main attributes of urban transport modes, measured on a 5-point Likert scale and differentiated by gender, are illustrated in Figure 4. The findings indicate that both women and

men consistently perceive the private car as a safe, time-efficient, and relatively secure mode of transport, but expensive. Motorcycles are viewed as high-risk modes, but are positively rated for their affordability and short travel times. For both genders, formal public transport is regarded as an affordable option, although it is also associated with long travel times and high levels of pollution (The bus fleet is mainly based on oil fuel energies) and personal security risks. In particular, women perceive public transport to be slightly more comfortable and faster than men, indicating a moderate gender disparity in the perceived travel experience. This difference may reflect adaptive perceptions shaped by greater reliance on these modes, rather than intrinsic preferences. We found that, from persons with a high perception of insecurity in public transit, 51.30% corresponds with women vs 48.70% for men. Similar results are found for motorcycles (52.33% women vs 47.67% men), and cars (54.30% women vs 45.70% men). Prior research shows that perceptions of safety, comfort, and reliability play a central role in shaping women's transport choices, especially in contexts where concerns about harassment and insecurity are prevalent (Vanderschuren et al., 2019; Allen, 2018). In this sense, attitudes toward transport modes are not formed in isolation but are closely linked to lived mobility constraints. Gendered experiences in the transport system influence how attributes are evaluated, reinforcing observed differences in modal choice and access.

3.2. Determinants of Modal Choice

3.2.1. Individual characteristics

Table 5 displays the estimated coefficients and odds ratios (OR) of the MNL model analyzing the choice of transport mode by gender. The findings of the pooled sample reveal significant gender differences in travel preferences. Specifically, women are 60% less likely to choose a car and 50% less likely to select a motorcycle compared to men, indicating a stronger reliance on public transportation among women. To further explore these differences, the MNL model was also estimated separately for women and men subsamples (see columns [2] and [3] of Table 5).

Our findings indicate that age significantly influences transport mode choice, with notable gender differences. Specifically, men aged 35–54 are more likely to drive a car than use public transport (OR = 2.69). This trend may reflect the transition from youth to middle age, which reduces demand for public transport and increases preference for private modes such as cars, potentially facilitated by higher incomes (Golob and Hensher, 2007; Basarić et al., 2016). In contrast, older men (≥ 55) are less likely to use motorcycles than public transport compared to those aged 20–34 (OR = 0.31), corresponding to a 69% reduction in probability. A similar pattern is observed among women, who exhibit a higher reduction in the probability of choosing motorcycles over public transport in older age groups (OR = 0.16).

These trends may be associated to age-related declines in mobility (with less frequent and shorter trips), as well

as increased reliance on public transport and car-passenger roles, particularly among women whose mobility is often more restricted (Rosenbloom, 2006; Golob and Hensher, 2007). Additionally, both young men and women (aged 15–19 years) are considerably less likely to use motorcycles than public transport compared to those aged 20–34 (OR = 0.14 and OR = 0.29, respectively), potentially reflecting licensing restrictions (65.17% of men versus 40.51% of women are licensed drivers in the sample) and affordability constraints.

Analysis of education level shows that individuals with tertiary education are more likely to use private vehicles—particularly cars (OR = 1.76) and motorcycles (OR = 1.04)—instead of public transport (Table 5). This effect is especially pronounced among men: those with higher education (tertiary) are substantially more likely to choose a car (OR = 2.58), while those with only basic education (primary) are significantly less likely to do so (OR = 0.35). For women, tertiary education increases the probability choosing a car and motorcycles with OR 1.51 and 1.08 respectively against women with secondary education level. These patterns suggest that, while education—which is closely linked to higher income and occupational status—facilitates access to private vehicles, this effect is uneven across genders.

Gender asymmetry is not merely a result of individual endowments but also reflects structural barriers. Men are more likely to access private vehicles, such as cars (Dobbs, 2005; Vance and Iovanna, 2007), and prior research shows that men are more likely to exercise control over mobility resources, such as household vehicles. Consequently, women may face restricted and mediated access to private vehicles, reflecting intra-household disparities (Scheiner and Holz-Rau, 2012; Dobbs, 2007; Tilley and Houston, 2016). Such asymmetries in access can limit women's ability to translate educational attainment into improved transport autonomy.

In the same sense, the intersection of occupational status and gender exerts a significant influence on transport mode choices. Among women, we found that, compared to wage earners and self-employed individuals, domestic workers are 89% less likely to use cars (OR = 0.11) and 77% less likely to use motorcycles (OR = 0.23), underscoring substantial barriers to private transportation access. This aligns with existing evidence that women's travel is constrained by disproportionate caregiving and household responsibilities (Steiner, 1998; Craig and van Tienoven, 2019). These constraints are also manifest spatially, limiting distance women can travel, the number of trips, and often confining them to part-time or low-wage jobs located near the home (Kim, Sang, Chun and Lee, 2012; de Meester and van Ham, 2009), especially among less-educated women (Havet et al., 2021).

3.2.2. Travel patterns

Travel distance emerges as a key determinant of transport mode choice, with distinct gendered patterns. The probability of choosing a car over public transport increases with travel distance, particularly for trips between 11–20 km.

In contrast, motorcycle use declines notably for the same distance range, with a 50% reduction in the odds of choosing a motorcycle compared to shorter trips (≤ 5 km), indicating that motorcycles are perceived as less suitable for longer distances as found in other studies (S. and Chalumuri (Ch.), 2016; Havet et al., 2021; Goel, 2023).

Gender differences in distance-related mode choice are pronounced. Among men, medium-distance trips (6–20 km) are strongly associated with increased car use, indicating a marked dependence on private vehicles for longer commutes. However, among women, longer distances do not significantly increase the odds of choosing a car, highlighting additional barriers or constraints in accessing private transportation as has been addressed in previous research of Adeel et al. (2017) and Basbas et al. (2023).

Trip purpose also significantly influences transport mode choice and reveals additional gendered patterns. Compared to work-related travel, non-work trips decrease the probability of choosing private vehicles. Notably, the general pattern of women preferring public over private transportation (in the pooled model) does not hold when trip purposes involve caregiving, or shopping/errands as shown in the model with the women subsample. For these specific trip types, women exhibit an increased probability of choosing cars and a decreased probability of choosing motorcycles, likely reflecting safety concerns associated with transporting dependents or carrying goods which is in line with Fan (2017), Srinivasan (2005) and Dobbs (2005).

Regarding trip chaining, measured by the number of stops, the pooled model reveals that individuals making two or more stops have a decreased probability of using private vehicles compared to public transport. Specifically, comparing users with two or more trip stops against users without stops, the probability of using cars decreases by 48% (OR = 0.52) and motorcycle use decreases by 40% (OR = 0.60). While trip chaining does not significantly affect car use among men and may slightly increase motorcycle use, the effect among women is particularly pronounced, i.e., there is more probability to use car than public transport by women whose travel has 1 stop (OR=1.25). This result aligns with existing literature on the complexities of trip chaining for women, who face additional constraints when combining multiple activities in a single journey (McGuckin et al., 2005; Scheiner and Holz-Rau, 2017; Dowling, 2000; Ortiz Escalante, Ciocchetto, Fonseca, Casanovas and Valdivia, 2021). Women who make multiple stops have a 66.6% lower probability of using a car (OR = 0.33) and a 57.4% lower probability of using a motorcycle (OR = 0.43), indicating greater dependence on public transport for women with more complex trip patterns.

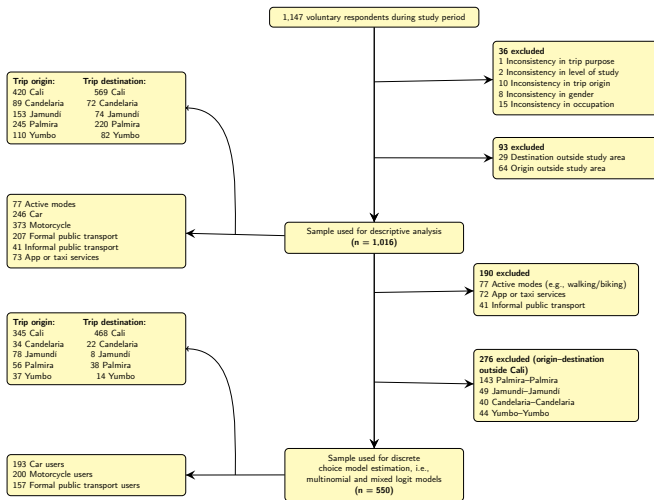


Figure 1: Flowchart of participants' selection

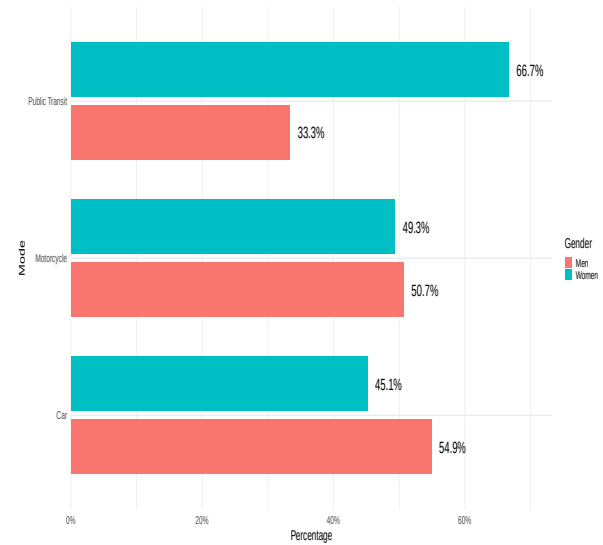


Figure 3: Use of transport modes by gender.

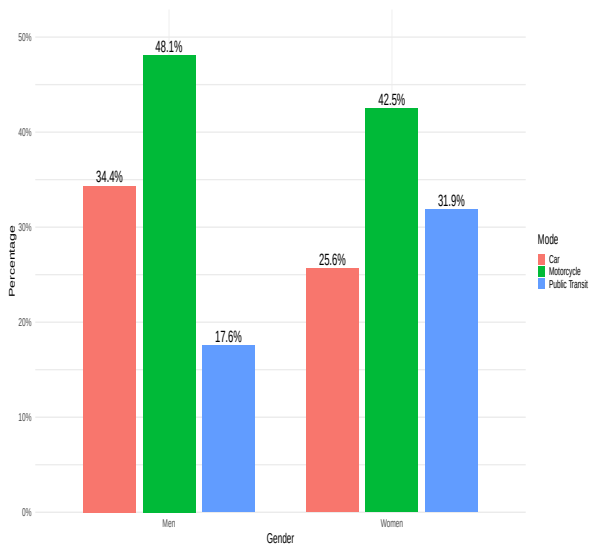


Figure 2: Gender use by transport modes.

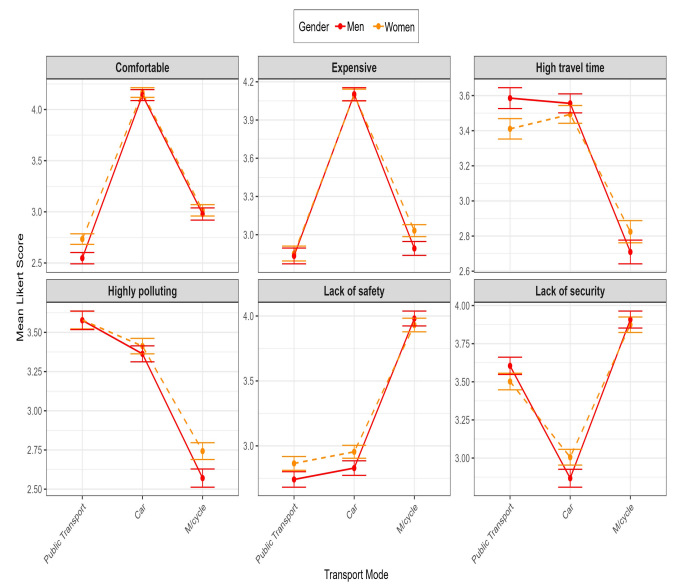


Figure 4: Mean Likert Score by Transport mode and Gender.

Table 5: Multinomial logit model estimates and odds ratio of transport mode choice by gender

	(1) Pooled			(2) Women			(3) Men		
	Car		M/cycle	Car		M/cycle	Car		M/cycle
	Coef.	Odds		Coef.	Odds		Coef.	Odds	
Intercept	-5.54 (1.09)***	0.00 (0.00-0.03)***	0.83 (0.70)	-5.63 (1.56)***	0.00 (0.00-0.08)***	1.00 (0.18-5.62)	-6.80 (1.77)***	0.00 (0.00-0.04)***	2.18 (0.21-22.53)
Distance (km)									
6-10	0.17 (0.43)	1.18 (0.51-2.75)	-0.23 (0.39)	0.06 (0.58)	1.06 (0.34-3.32)	0.87 (0.31-2.44)	0.12 (0.72)	1.13 (0.27-4.67)	0.51 (0.14-1.83)
11-20	0.26 (0.41)	1.29 (0.57-2.90)	-0.69 (0.38)*	-0.61 (0.56)	0.50 (0.18-1.62)	0.52 (0.20-1.34)	1.55 (0.74)**	4.69 (1.10-20.00)**	0.44 (0.12-1.66)
>20	0.01 (0.49)	1.01 (0.38-2.64)	-0.53 (0.44)	-0.62 (0.68)	0.54 (0.14-2.04)	0.57 (0.18-1.81)	0.68 (0.84)	1.97 (0.38-10.29)	0.46 (0.10-2.05)
Age (years)									
15-19	-0.47 (0.44)	0.63 (0.26-1.48)	-1.46 (0.50)***	-0.31 (0.61)	0.73 (0.22-2.42)	0.29 (0.08-1.13)*	-0.69 (0.74)	0.50 (0.12-2.13)	0.14 (0.03-0.65)***
35-54	0.63 (0.39)*	1.88 (0.88-4.01)*	-0.25 (0.35)	0.27 (0.54)	1.31 (0.45-3.81)	0.84 (0.35-2.06)	0.99 (0.67)	2.69 (0.73-9.91)	0.77 (0.23-2.55)
≥55	-0.51 (0.50)	0.60 (0.23-1.60)	-1.29 (0.45)***	0.32 (0.76)	1.38 (0.31-6.06)	0.16 (0.04-0.67)***	-1.41 (0.78)*	0.24 (0.05-1.13)*	0.31 (0.08-1.16)*
Gender									
Women	-0.90 (0.28)***	0.40 (0.23-0.70)***	-0.69 (0.26)***	—	—	—	—	—	—
Education									
Primary	-0.59 (0.65)	0.55 (0.15-2.00)	-0.25 (0.54)	-0.21 (0.94)	0.81 (0.13-5.09)	1.65 (0.40-6.81)	-1.05 (1.04)	0.35 (0.05-2.68)	0.29 (0.06-1.36)
Tertiary	0.56 (0.30)*	1.76 (0.98-3.14)*	0.04 (0.28)	0.41 (0.41)	1.51 (0.68-3.35)	1.08 (0.53-2.23)	0.95 (0.52)*	2.58 (0.93-7.18)*	1.10 (0.42-2.84)
Occupation									
Domestic work	-1.63 (0.91)*	0.20 (0.03-1.17)*	-1.40 (0.72)*	-2.17 (1.01)**	0.11 (0.02-0.83)**	0.23 (0.05-1.10)*	—	—	—
Unemployed / inactive	-0.36 (0.68)	0.70 (0.18-2.65)	-0.31 (0.59)	-1.40 (1.22)	0.25 (0.02-2.68)	0.93 (0.18-4.93)	1.17 (1.09)	3.23 (0.38-27.40)	1.25 (0.19-8.19)
Student	-0.01 (0.53)	0.99 (0.35-2.82)	-0.47 (0.46)	-0.01 (0.80)	0.99 (0.21-4.78)	0.60 (0.17-2.16)	0.07 (0.81)	1.07 (0.22-5.23)	0.82 (0.21-3.24)
Purpose of travel									
Shopping	-0.12 (0.71)	0.89 (0.22-3.59)	0.05 (0.64)	-0.48 (1.21)	0.62 (0.06-6.67)	1.82 (0.27-12.18)	0.02 (1.14)	1.02 (0.11-9.49)	0.76 (0.11-5.17)
Caregiving/family	-0.21 (0.80)	0.81 (0.17-3.88)	-0.43 (0.80)	0.21 (1.29)	1.23 (0.10-15.49)	0.43 (0.03-6.70)	-0.12 (1.18)	0.89 (0.09-8.95)	0.46 (0.05-3.89)
Education	-0.03 (0.53)	0.97 (0.34-2.75)	-0.81 (0.49)	-0.08 (0.76)	0.92 (0.21-4.12)	0.63 (0.16-2.52)	-0.16 (0.88)	0.85 (0.15-4.77)	0.26 (0.06-1.17)*
Recreation/health	-0.79 (0.51)	0.45 (0.17-1.23)	-0.81 (0.45)	-0.91 (0.65)	0.40 (0.11-1.45)	0.53 (0.17-1.67)	-1.13 (0.96)	0.32 (0.05-2.12)	0.39 (0.08-1.86)
Number of stops									
1 stop	-0.14 (0.38)	0.87 (0.41-1.85)	-0.40 (0.38)	0.23 (0.55)	1.25 (0.42-3.72)	0.68 (0.23-2.07)	-0.62 (0.60)	0.54 (0.17-1.75)	0.59 (0.19-1.84)
≥2 stops	-0.65 (0.35)*	0.52 (0.26-1.03)*	-0.51 (0.32)	-1.10 (0.48)**	0.33 (0.13-0.85)**	0.43 (0.19-0.96)**	-0.15 (0.63)	0.86 (0.25-2.96)	1.04 (0.33-3.30)
Municipality									
Call	-0.58 (0.33)*	0.56 (0.29-1.07)*	-0.27 (0.29)	-0.75 (0.45)*	0.47 (0.20-1.14)*	1.02 (0.48-2.17)	-0.69 (0.57)	0.50 (0.16-1.55)	0.50 (0.16-1.30)

	Coef	Odds Ratio	Coef	Odds Ratio	Coef	Odds Ratio
Cost/Wage	-0.36 (0.11)***	0.70 (0.56-0.87)***	-0.24 (0.15)*	0.79 (0.59-1.05)*	-0.67 (0.21)***	0.51 (0.34-0.78)***
Safety	-0.04 (0.02)**	0.96 (0.93-0.99)**	-0.02 (0.02)	0.98 (0.94-1.02)	-0.07 (0.03)**	0.94 (0.89-0.98)**
Security	-0.04 (0.01)***	0.96 (0.93-0.99)***	-0.03 (0.02)	0.97 (0.93-1.01)	-0.07 (0.02)***	0.93 (0.89-0.98)***
N	544			280		264
ll(model)	-447.49			-236.17		-185.76
AIC	988.98			562.33		457.52
BIC	1191.03			725.90		611.29

Wald-Test: * p-value < 0.05, ** p-value < 0.01, *** p-value < 0.001

3.3. Sensitivity of Modal Choice

The MXL model incorporates random coefficients for the cost-to-income ratio, safety relevance factor, and security relevance factor with each coefficient defined to follow a zero-bounded triangle distribution to ensure theoretically consistent negative signs. Figure 5 displays the density estimates of the simulated random parameters across three specifications: the pooled sample and two gender-disaggregated models. The distributions highlight significant gender-based heterogeneity, especially in the safety relevance factor.

3.3.1. Sensitivity of Mode Choice to Travel Cost-to-Income Changes

Table 6 reports the own- and cross-elasticities with respect to the cost-to-income ratio, which measures how changes in travel affordability influence the probability of mode choice. The own-elasticities (diagonal values) are positive and less than one across all modes, indicating inelastic responses. This suggests that, although affordability is a relevant determinant of mode choice, changes in this attribute alone may not be sufficient to induce substantial shifts in urban travel behavior. However, variation between own elasticities reveals differences by transport mode: car users show the highest sensitivity (0.48%), followed by public transport users (0.32%), while motorcycle users are the least responsive (0.13%). Cross-elasticities (off-diagonal elements) are negative, indicating substitution across transport modes in response to changes in relative affordability. Specifically, a decrease in car travel costs results in a slight reduction in the probability of choosing motorcycles (-0.22%) and public transport (-0.21%). Similarly, improved public transport affordability reduces the probability of choosing motorcycles (-0.14%) and, to a lesser extent, cars (-0.12%), potentially reflecting cost-related barriers to shifting from motorcycles to cars.

When analyzing travel behavior by gender, some interesting differences emerge in how men and women respond to changes in transportation costs. Overall, neither group dramatically changes their transport choices when prices shift slightly—a pattern known as inelastic behavior—but the degree of sensitivity does vary between genders. When costs change, women and men do not respond equally. Women are more sensitive than men to price changes in private transport modes. If using a car becomes more expensive, women are more likely to abandon it (0.52% reduction in probability vs. 0.46% for men). The same holds for motorcycles: women react more strongly to cost increases (0.16% vs. 0.12%). This suggests that women operate with tighter financial constraints when choosing private transport, making cost a more decisive barrier for them than for men.

The picture reverses for public transport: men are more responsive to changes in public transit affordability (0.40% vs. 0.26% for women), suggesting that men consider public transport a more viable alternative when it becomes cheaper relative to driving. Among women who already use private vehicles, however, this response is notably weaker — even

Table 6

Direct and Cross Marginal Effects and Elasticities of Mode Choice by Gender

		Marginal Effects (MFEX)			Elasticities		
		1% Change in Cost-to-Income Ratio			1% Change in Cost-to-Income Ratio		
		Car	M/Cycle	Public Transport	Car	M/Cycle	Public Transport
Pooled	Car	0.00127	-0.00020	-0.00033	0.48074	-0.07270	-0.12274
	M/Cycle	-0.00073	0.00041	-0.00040	-0.21989	0.13452	-0.14206
	Public Transport	-0.00054	-0.00021	0.00073	-0.21507	-0.08529	0.31867
Women	Car	0.00115	-0.00014	-0.00038	0.52057	-0.05528	-0.17282
	M/Cycle	-0.00051	0.00036	-0.00045	-0.19038	0.15781	-0.20111
	Public Transport	-0.00065	-0.00022	0.00083	-0.18451	-0.06423	0.25568
Men	Car	0.00124	-0.00023	-0.00030	0.45960	-0.08588	-0.09564
	M/Cycle	-0.00081	0.00041	-0.00037	-0.24607	0.12347	-0.11015
	Public Transport	-0.00043	-0.00018	0.00067	-0.24480	-0.10139	0.39929

Note: Marginal effects (MFEX) represent changes in choice probability (in percentage points) in response to one increase in the cost-to-income ratio. Elasticities represent the percentage change in choice probability in response to an increase of 1% in the cost-to-income ratio.

when public transport becomes more affordable, they are less inclined to switch away from their current mode than men. This suggests that cost alone does not fully explain women's transport decisions. Women who use private vehicles appear to value additional attributes these modes provide — such as safety, flexibility, comfort, or a sense of autonomy — that public transport may not yet offer to the same extent. In other words, making public transport cheaper may not be sufficient on its own to attract women away from private vehicles, consistent with evidence suggesting that subsidies alone have heterogeneous effects on ridership depending on user profiles as mentioned in previous studies Guzman and Cantillo-Garcia (2024).

3.3.2. Sensitivity of Mode Choice to Improvements in Personal Security

In the context of developing countries, personal security is an important concern that reflects perceptions of personal safety regarding crime, theft, and harassment associated with various modes of transport. Especially for women, this can be an influential factor in shifting towards private modes such as motorcycles (Abonado, 2023). This is one of the aspects related to the decline in public transit in many countries around the world, specifically in Latin America (Noguera, 2017).

Table 7 presents both own- and cross-elasticities of transport mode choice in response to changes in the personal security relevance factor. Own-elasticities indicate that public transport shows the highest responsiveness to security improvements in the pooled (0.97%) and men (1.23%) samples, while women demonstrate the greatest sensitivity to motorcycles (0.82%), followed by public transport (0.76%) and cars (0.68%).

Cross-elasticities are consistently negative, indicating substitution effects across modes in response to enhanced security. Greater car security reduces the probability of choosing motorcycles and public transport, with effects more pronounced for men (-0.44% and -0.46%, respectively) than women (-0.27% and -0.27%). A similar trend is observed

Table 7

Direct and Cross Elasticities of Mode Choice by Gender in Response to a Positive Change in Security

		Marginal Effects (MFEX)			Elasticities		
		Change in Security = +1			1% Change in Security		
		Car	M/Cycle	Public Transport	Car	M/Cycle	Public Transport
Pooled	Car	0.0179	-0.0104	-0.0076	0.6804	-0.4452	-0.3730
	M/Cycle	-0.0104	0.0178	-0.0075	-0.3428	0.8496	-0.3625
	Public Transport	-0.0075	-0.0074	0.0151	-0.3431	-0.4341	0.9723
Women	Car	0.0147	-0.0070	-0.0077	0.6849	-0.3488	-0.4853
	M/Cycle	-0.0070	0.0149	-0.0079	-0.2741	0.8254	-0.4778
	Public Transport	-0.0077	-0.0079	0.0156	-0.2714	-0.3390	0.7636
Men	Car	0.0197	-0.0124	-0.0075	0.7271	-0.5597	-0.3039
	M/Cycle	-0.0124	0.0193	-0.0071	-0.4429	0.9760	-0.2884
	Public Transport	-0.0073	-0.0069	0.0145	-0.4599	-0.5502	1.2296

Note: Marginal effects (MFEX) represent changes in choice probability (in percentage points) in response to a unit change in security by mode of transport, while elasticities represent the percentage change in choice probability in response to a 1% change in security.

for motorcycles, where increased security is associated with a greater decrease in the probability of choosing cars and public transport among men (-0.56% and -0.55%, respectively) relative to women (-0.35% and -0.34%, respectively). Notably, the cross-elasticities for public transport exhibit the opposite pattern: women display stronger substitution effects (-0.49% for car and -0.48% for motorcycle) compared to men (-0.30% and -0.29%). This result suggests that improvements in public transport security may encourage female users to shift from private modes to public options, reinforcing the critical role of security in promoting gender-equitable mobility choices.

3.3.3. Sensitivity of Mode Choice to Improvements in Road Traffic Safety

High road accident rates are a public health problem worldwide, especially in motorcycle-dependent cities (Caldavid and Salazar-Serna, 2021). Against this backdrop, we seek to understand whether actions designed to enhance the perceived road safety of transport modes can generate changes in modal choice behavior, and whether such effects differ between men and women.

The own- and cross-elasticities of transport mode choice in response to changes in the safety relevance factor are detailed in Table 8. These results capture users' perceptions of safety, encompassing aspects such as road security, accident risk, and related concerns. Motorcycles, recognized as a high-risk mode of transport, exhibit the highest responsiveness to safety improvements, with an own-elasticity of 0.46%, followed by public transport (0.36%) and cars (0.33%). This trend is consistent across genders; however, men users display notably higher responsiveness compared to women users, particularly in the case of motorcycles (0.66% vs. 0.28%).

These findings underscore heightened concern regarding motorcycle safety, which can be attributed to their increased vulnerability in traffic and their association with higher rates of accidents and fatalities.

Regarding substitution effects, improvements in car safety reduce the probability of choosing public transport and, to a lesser extent, motorcycles. These effects are more

Table 8

Direct and Cross Elasticities of Mode Choice by Gender in Response to a Positive Change in Safety

		Marginal Effects (MFEX)			Elasticities		
		1% Change in safety			1% Change in safety		
		Car	M/Cycle	Public Transport	Car	M/Cycle	Public Transport
Pooled	Car	0.00988	-0.00564	-0.00425	0.33279	-0.22772	-0.13865
	M/Cycle	-0.00564	0.00984	-0.00422	-0.15515	0.45908	-0.13610
	Public Transport	-0.00424	-0.00421	0.00848	-0.16387	-0.23558	0.34953
Women	Car	0.00487	-0.00228	-0.00258	0.20473	-0.10904	-0.10849
	M/Cycle	-0.00228	0.00496	-0.00267	-0.07963	0.28263	-0.10847
	Public Transport	-0.00258	-0.00268	0.00525	-0.08199	-0.11202	0.16659
Men	Car	0.01445	-0.00897	-0.00557	0.42956	-0.35764	-0.13295
	M/Cycle	-0.00897	0.01417	-0.00527	-0.24366	0.65766	-0.12723
	Public Transport	-0.00548	-0.00519	0.01084	-0.26774	-0.37520	0.57010

Note: Marginal effects (MFEX) represent changes in choice probability (in percentage points) in response to a 1% change in safety, while elasticities represent the percentage change in choice probability relative to a 1% change in safety.

pronounced among men (-0.27% for public transport and -0.24% for motorcycles) than among women (-0.08% and -0.08%, respectively). A similar pattern emerges for motorcycles: enhancements in motorcycle safety primarily decrease the choice of public transport (-0.24%) and, to a lesser degree, cars (-0.23%), with stronger responses observed among men (-0.37% for public transport and -0.36% for cars) compared to women (-0.11% and 0.11%, respectively).

These substitution patterns may reflect perceived trade-offs between safety and other factors such as affordability, convenience, and flexibility —particularly for individuals who perceive motorcycles as an accessible yet risk-prone alternative—. In contrast, gender differences in the response to improvements in public transport safety are less substantial; enhancements in public transport safety lead to similar reductions in probability of choosing cars and motorcycles across both genders, although the effect remains slightly more pronounced among men users.

4. Conclusions

Transport policies have traditionally been implemented without considering gender differences; yet the challenge of building more inclusive and equitable urban mobility systems demands that the impacts of such policies be recognized and analyzed in a differentiated way. This paper contributes to the evidence base needed for designing gender-responsive urban transport policies in developing countries. Using a Mixed Logit model applied to survey data from Cali, Colombia — a city emblematic of mobility challenges in the Global South — we conducted a gender-disaggregated sensitivity analysis of transport mode choice. Our findings reveal that transport policies do not affect men and women equally: the direction, magnitude, and nature of behavioral responses to changes in cost, personal security, and road safety differ substantially by gender. These results carry direct implications for how policymakers design and evaluate transport interventions, particularly in contexts where public transit decline, motorcycle dependence, and gender-based insecurity intersect.

4.1. Key findings and their policy relevance

Perhaps the most policy-relevant finding of this study is the outsized role of perceived personal security in shaping women's modal choices. Women exhibit greater sensitivity than men to improvements in public transport security (own-elasticity of 0.76% vs. 1.23% for men in absolute terms; but with stronger cross-substitution effects of -0.49% from private modes), suggesting that security-enhancing interventions in public transport are more likely to attract women away from private vehicles than cost reductions alone. This finding is consistent with evidence from Cali's own experience: the security strategy implemented by the city's MIO system in 2024 — encompassing user reporting mechanisms, increased enforcement presence in bus stations, surveillance in exclusive transit lanes, and public space improvements — yielded a 60% reduction in theft incidents and a 10% increase in ridership (Alcaldía de Santiago de Cali, 2024, 2025). These real-world outcomes lend further credibility to our elasticity estimates and underscore that security investment in public transport is not merely a service quality improvement — it is an equity intervention.

Although travel behavior is inelastic with respect to cost-to-income changes across all groups, women are consistently more responsive than men to price increases in private modes — particularly cars (0.52% vs. 0.46%) and motorcycles (0.16% vs. 0.12%). This reflects the tighter financial constraints under which women operate, driven by the structural inequalities documented in our descriptive analysis: women in the sample are overrepresented in lower socioeconomic strata, face higher rates of unemployment, and bear the burden of unpaid domestic work, all of which constrain their transport budgets. Conversely, men are more sensitive to public transport affordability improvements (0.40% vs. 0.26%), suggesting that fare incentives in public transit are more effective at attracting male users.

Policy implication: Fare subsidy programs and cost reduction policies should be differentiated by gender and income level. Targeted subsidies or income-based fare structures for public transit would be more effective at addressing the mobility constraints of low-income women than uniform fare reductions. Policymakers should avoid designing affordability interventions as gender-neutral when the underlying constraints are clearly gendered.

Men show substantially higher sensitivity to road safety improvements, especially for motorcycles (0.66% vs. 0.28% for women). This asymmetry reflects a critical distinction: for men, "safety" is primarily understood in terms of crash risk and traffic infrastructure, whereas for women, the perceived risk encompasses personal insecurity, harassment, and vulnerability in public spaces. This distinction means that road safety campaigns and infrastructure investments — however important — will have limited effectiveness in improving women's public transport uptake unless personal security concerns are simultaneously addressed.

Policy implication: Road safety policies should be disaggregated by gender in both design and evaluation. Campaigns targeting motorcycle safety risk reduction are likely

to generate stronger behavioral responses among men. For women, integrated approaches combining road safety with personal security, station design, and anti-harassment enforcement will be more impactful.

Beyond attribute sensitivity, the determinants analysis reveals that women's transport choices are significantly constrained by structural factors: occupational status (domestic workers face the greatest barriers to private transport access), trip chaining complexity, and limited access to mobility resources. Women with higher educational attainment do not necessarily translate that advantage into improved transport autonomy — a finding that highlights how structural and intra-household inequalities mediate the relationship between socioeconomic status and mobility.

Policy implication: Gender-responsive transport planning must go beyond infrastructure and pricing. Policies that address women's time poverty — such as improved multimodal trip-chaining options, childcare-accessible transit schedules, and land-use planning that reduces travel distances — are essential complements to security and affordability interventions.

4.2. Recommendations for policy and future research

The findings of this study point to the following actionable recommendations for transport policymakers and planners in developing-country cities:

Adopt gender-disaggregated impact assessment as a standard practice in transport policy evaluation. The sensitivity analysis framework presented here — combining MXL models with direct and cross-elasticities by gender — provides a replicable and rigorous tool for assessing whether proposed interventions will reduce or reinforce mobility inequalities. Invest in personal security as a core public transport strategy, not merely as a complementary measure. The evidence suggests that security improvements have the greatest potential to shift women's modal choices toward public transit, with multiplier effects on ridership and system viability.

Design targeted affordability mechanisms that account for gender and income disparities. Means-tested fare structures or mobility subsidies for low-income women would be more effective than universal fare reductions in promoting equitable access.

Engage with motorcycle governance as a gender issue. The masculinization of motorcycle use restricts women's autonomous mobility and exposes them to greater safety risks. Policies regulating motorcycle use, improving safety standards, and expanding public transit coverage in motorcycle-dependent municipalities must account for gendered impacts.

Investigate the gender implications of emerging mobility trends — including ride-hailing and micromobility — to assess whether these innovations are alleviating or reproducing the barriers identified in this study.

Acknowledgement

This research was supported by the WWB Foundation Fund, grant [190-2023]; and the Ministry of Science, Technology and Innovation (Minciencias) of Colombia through the *Women in Artificial Intelligence, Science, and Quantum Technologies 2025* program, grant [112721-200-2025].

AI use statetment

During the preparation of this work, the authors, as non-native English speakers, used Claude Sonnet 4.6 for content organization and to improve language and readability. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

References

- Abonado, F., 2023. Women in colombia seek safety on motorbikes – dw. deutsche welle- society colombia. URL: <https://www.dw.com/en/women-in-colombia-seek-safety-on-motorbikes/video-66735713>.
- Adeel, M., Yeh, A.G., Zhang, F., 2017. Gender inequality in mobility and mode choice in pakistan. *Transportation* 44, 1519–1534. doi:10.1007/s11116-016-9712-8.
- Adeel, M., Yeh, A.G.O., Zhang, F., 2016. Gender and transportation in developing megacities: A case study of karachi, pakistan. *Transportation Research Part A: Policy and Practice* 94, 95–107. doi:10.1016/j.tra.2016.08.024.
- Agencia Nacional de Seguridad Vial, 2025. Observatorio-estadísticas. URL: <https://ansv.gov.co/observatorio/estad%C3%ADsticas>.
- Alcaldía de Santiago de Cali, 2024. Con buenos resultados avanza la estrategia de seguridad en el sistema integrado de transporte masivo, mio. URL: <https://www.cali.gov.co/seguridad/publicaciones/180289/con-buenos-resultados-avanza-la-estrategia-de-seguridad-en-el-sistema-integrado-de-transporte-masivo-mio/>.
- Alcaldía de Santiago de Cali, 2025. Más de 8 millones de viajes adicionales, mejoría en el cumplimiento y mayor satisfacción de los usuarios, entre logros del mio en 2024. URL: <https://www.cali.gov.co/movilidad/publicaciones/184390/mas-de-8-millones-de-viajes-adicionales-mejoria-en-el-cumplimiento-y-mayor-satisfaccion-de-los-usuarios-entre-logros-del-mio-en-2024/>.
- Allen, H., 2018. Ella se mueve segura.
- Asociación Nacional de Empresarios de Colombia, 2024. Nuevo estudio de las motocicletas en colombia: el 90% de los hogares con motocicletas en colombia pertenece a estratos bajos. URL: <https://www.andi.com.co/Home/Noticia/17736-nuevo-estudio-de-las-motocicletas-en-co>.
- Basarić, V., Vujičić, A., Simić, J.M., Bogdanović, V., Saulić, N., 2016. Gender and age differences in the travel behavior - a novi sad case study. *Transportation Research Procedia* 14, 4324–4333. doi:10.1016/j.trpro.2016.05.354.
- Basbas, S., Papagiannakis, A., Baraklianos, I., Nikiforiadis, A., Campisi, T., 2023. Does gender matter in daily urban mobility? Exploring travel perceptions, attitudes, and behaviours. *International Journal of Transport Development and Integration* 7, 67–75. doi:10.18280/ijtdi.070201.
- Cadavid, L., Salazar-Serna, K., 2021. Mapping the research landscape for the motorcycle market policies: Sustainability as a trend—a systematic literature review. *Sustainability* 13. URL: <https://www.mdpi.com/2071-1050/13/19/10813>, doi:10.3390/su131910813.
- Cao, X., Mokhtarian, P.L., Handy, S.L., 2008. Differentiating the influence of accessibility, attitudes, and demographics on stop participation and frequency during the evening commute. *Environment and Planning B: Planning and Design* 35, 431–442. doi:10.1068/b32056.
- Caracol Radio, 2023. ¿a cuántos kilómetros es conveniente cambiar un carro o una moto? URL: <https://caracol.com.co/2023/10/06/a-cuantos-kilometros-es-conveniente-cambiar-un-carro-o-una-moto/>.
- Carro Ya, 2023. Motos, patinetas y bicicletas. URL: <https://www.carroya.com/motos>.
- CEPAL, 2019. Determinantes de género en las políticas de movilidad urbana en América Latina. Technical Report. CEPAL. URL: <https://repositorio.cepal.org/server/api/core/bitstreams/16d60ffe-7c50-4cf8-aad1-fc56aab7e11e/content>.
- Craig, L., van Tienoven, T.P., 2019. Gender, mobility and parental shares of daily travel with and for children: a cross-national time use comparison. *Journal of Transport Geography* 76, 93–102. doi:10.1016/j.jtrangeo.2019.03.006.
- Cuéllar-Álvarez, Y., Clappier, A., Rojas-Roa, N.Y., Thunis, P., Mangones, S., Belalcázar-Cerón, L.C., 2024. Emissions and ownership-cost of conventional and electric passenger vehicles in bogotá, colombia. *Transportation Research Part D: Transport and Environment* 128, 104105. URL: <https://www.sciencedirect.com/science/article/pii/S1361920924000622>, doi:https://doi.org/10.1016/j.trd.2024.104105.
- DANE, 2018. Censo nacional población y vivienda. <https://www.dane.gov.co/index.php/estadisticas-por-tema/demografia-y-poblacion/proyecciones-de-poblacion>.
- Dobbs, L., 2005. Wedded to the car: Women, employment and the importance of private transport. *Transport Policy* 12, 266–278. doi:10.1016/j.tranpol.2005.02.004.
- Dobbs, L., 2007. Stuck in the slow lane: Reconceptualizing the links between gender, transport and employment. URL: <https://doi.org/10.1111/j.1468-0432.2007.00334.x>.
- Domencich, T.A., McFadden, D., 1975. *Urban Travel Demand: A Behavioral Analysis: a Charles River Associates Research Study*, Volumen 93. North-Holland Publishing Company.
- Dowling, R., 2000. Cultures of mothering and car use in suburban sydney: a preliminary investigation. *Geoforum* 31, 345–353. URL: [https://doi.org/10.1016/S0016-7185\(99\)00048-2](https://doi.org/10.1016/S0016-7185(99)00048-2).
- El Espectador, 2024a. Comprar carro de segunda: ¿cómo identificar si el kilometraje está alterado? URL: <https://www.elespectador.com/autos/comprar-carro-de-segunda-como-identificar-si-el-kilometraje-esta-alterado/>.
- El Espectador, 2024b. ¿cuál es la vida útil de un carro, según años y kilometraje? URL: <https://www.elespectador.com/autos/cual-es-la-vida-util-de-un-carro-segun-anos-y-kilometraje/>.
- El Ministerio de Hacienda y Crédito Público, 2023. Decreto 2228 de 2023. URL: <https://www.alcaldiabogota.gov.co/sisjur/normas/Normal.jsp?i=152439&dt=S>.
- El País, 2023. ¿por qué las pólizas de seguro para los carros están por las nubes? estas son las razones de los altos precios. URL: <https://www.elpais.com.co/economia/por-que-las-polizas-de-seguro-para-los-carros-estan-por-las-nubes-estas-son-las-razones-de-los-altos-precios.html>.
- El Tiempo, 2022. ¿cada cuánto tiempo los colombianos cambian de carro? URL: <https://www.eltiempo.com/economia/finanzas-personales/cada-cuanto-se-cambia-de-carro-en-colombia-698426>.
- El Tiempo, 2024a. Carros: hasta cuántos kilómetros dura el motor y cuándo conviene venderlo. URL: <https://www.eltiempo.com/cultura/gente/carros-hasta-cuantos-kilometros-dura-el-motor-y-cuando-conviene-venderlo-845050>.
- El Tiempo, 2024b. Este es el kilometraje ideal que debe tener un carro usado en colombia. URL: <https://www.eltiempo.com/economia/finanzas-personales/kilometraje-ideal-que-debe-tener-un-carro-de-segunda-784862>.
- Elias, W., Benjamin, J., Shifan, Y., 2015. Gender differences in activity and travel behavior in the arab world. *Transport Policy* 44, 19–27. doi:10.1016/j.tranpol.2015.07.001.
- Elias, W., Newmark, G.L., Shifan, Y., 2008. Gender and travel behavior in two arab communities in Israel. *Transportation Research Record*, 75–83doi:10.3141/2067-09.

- Fan, Y., 2017. Household structure and gender differences in travel time: spouse/partner presence, parenthood, and breadwinner status. *Transportation* 44, 271–291. doi:<https://doi.org/10.1007/s11116-015-9637-7>.
- Goel, R., 2023. Gender gap in mobility outside home in urban india. *Travel Behaviour and Society* 32. URL: <https://doi.org/10.1016/j.tbs.2023.01.004>.
- Golob, T.F., Hensher, D.A., 2007. The trip chaining activity of sydney residents: A cross-section assessment by age group with a focus on seniors. *Journal of Transport Geography* 15, 298–312. doi:[10.1016/j.jtrangeo.2006.09.005](https://doi.org/10.1016/j.jtrangeo.2006.09.005).
- Guzman, L.A., Cantillo-Garcia, V.A., 2024. Exploring the effects of public transport subsidies on satisfaction and ridership. *Research in Transportation Business Management* 56, 101168. URL: <https://www.sciencedirect.com/science/article/pii/S2210539524000701>, doi:<https://doi.org/10.1016/j.rtbm.2024.101168>.
- Hagen, J.X., Pardo, C.F., Valente, J.B., 2016. Motivations for motorcycle use for urban travel in latin america: A qualitative study. *Transport Policy* 49, 93–104. doi:[10.1016/j.tranpol.2016.04.010](https://doi.org/10.1016/j.tranpol.2016.04.010).
- Hanson, S., 2010. Gender and mobility: New approaches for informing sustainability. *Gender, Place and Culture* 17, 5–23. doi:[10.1080/09663690903498225](https://doi.org/10.1080/09663690903498225).
- Havet, N., Bayart, C., Bonnel, P., 2021. Why do gender differences in daily mobility behaviours persist among workers? *Transportation Research Part A: Policy and Practice* 145, 34–48. doi:[10.1016/j.tra.2020.12.016](https://doi.org/10.1016/j.tra.2020.12.016).
- Hensher, D.A., Greene, W.H., 2003. The mixed logit model: The state of practice. *Transportation* 30, 133–176. URL: <https://doi.org/10.1023/A:1022558715350>, doi:[10.1023/A:1022558715350](https://doi.org/10.1023/A:1022558715350).
- Hine, J., Mitchell, F., 2001. Better for everyone? travel experiences and transport exclusion. *Urban Studies* 38, 319–332. doi:[10.1080/00420980020018619](https://doi.org/10.1080/00420980020018619).
- Kim, C., Sang, S., Chun, Y., Lee, W., 2012. Exploring urban commuting imbalance by jobs and gender. *Applied Geography* 32, 532–545. doi:[10.1016/j.apgeog.2011.07.001](https://doi.org/10.1016/j.apgeog.2011.07.001).
- Kremer, M., Leino, J., Miguel, E., Zwane, A.P., 2011. Spring cleaning: Rural water impacts, valuation, and property rights institutions*. *The Quarterly Journal of Economics* 126, 145–205. URL: <https://doi.org/10.1093/qje/qjq010>, doi:[10.1093/qje/qjq010](https://doi.org/10.1093/qje/qjq010).
- Kwan, M.P., 2005. Gender differences in space-time constraints. *Area* 32, 145–156. doi:<https://doi.org/10.1111/j.1475-4762.2000.tb00125.x>.
- La Comisión de Regulación de Energía y Gas - CREG, 2023. Precios de combustibles líquidos. URL: <https://creg.gov.co/publicaciones/15565/precios-de-combustibles-liquidos/>.
- La República, 2018. Conozca cada cuánto debería cambiar su carro para que no pierda valor. URL: <https://www.larepublica.co/consumo/conozca-cada-cuanto-deberia-cambiar-su-carro-para-que-no-pierda-valor-2756791>}. text=Y%20agreg%C3%B3%20que%20un%20carro,sirve%20de%20punto%20de%20referencia%E2%80%9D.
- La República, 2024. En promedio, una persona cambia de motocicleta en colombia entre seis y ocho años. URL: <https://www.larepublica.co/empresas/cada-cuanto-cambian-los-colombianos-de-moto-3893181>.
- León, G., Miguel, E., 2017. Risky transportation choices and the value of a statistical life. *American Economic Journal: Applied Economics* 9, 202–228. doi:[10.1257/app.20160140](https://doi.org/10.1257/app.20160140).
- Li, H., Guensler, R., Ogle, J., 2004. Comparing Women's and Men's Morning Commute Trip Chaining in Atlanta, Georgia, by Using Instrumented Vehicle Activity Data. *Transportation Research Board of the National Academies*. volume 35. pp. 14–20.
- McFadden, D., 1974. The measurement of urban travel demand. *Journal of Public Economics* 3, 303–328. URL: <https://eml.berkeley.edu/reprints/mcfadden/measurement.pdf>.
- McGuckin, N., Zmund, J., Nakamoto, Y., 2005. Trip-chaining trends in the united states: Understanding travel behavior for policy making. *Transportation Research Record* 1917, 199–204. URL: <https://doi.org/10.1177/0361198105191700122>.
- de Meester, E., van Ham, M., 2009. Symmetry and asymmetry in working and commuting arrangements between partners in the netherlands: Does the residential context matter? *Environment and Planning A* 41, 2181–2200. doi:[10.1068/a41246](https://doi.org/10.1068/a41246).
- Mercado Libre, 2023. Motos. URL: <https://motos.mercadolibre.com.co/>.
- Myung-Jin, J., Ji-Eun, J., Hyun-Ju, A., 2018. The welfare effects of the free subway fare scheme for seniors: A discrete choice approach with the case of seoul. *Case Studies on Transport Policy* 6, 642–650. doi:[10.1016/j.cstp.2018.08.003](https://doi.org/10.1016/j.cstp.2018.08.003).
- Namgung, M., Akar, G., 2014. Role of gender and attitudes on public transportation use. *Transportation Research Record* 2415, 136–144. doi:[10.3141/2415-15](https://doi.org/10.3141/2415-15).
- Nasrin, S., Chowdhury, S., 2024. Exploring transport mobility issues and adaptive behavior of women in a developing country. *Transportation Research Interdisciplinary Perspectives* 23, 100991. URL: <https://www.sciencedirect.com/science/article/pii/S2590198223002385>, doi:<https://doi.org/10.1016/j.trip.2023.100991>.
- Ng, W.S., Acker, A., 2018. Understanding urban travel behaviour by gender for efficient and equitable transport policies. URL: <https://www.itf-oecd.org/sites/default/files/docs/urban-travel-behaviour-gender.pdf>.
- Noguera, I., 2017. Más mujeres en cali se mueven en moto por las calles de cali. URL: <https://www.eltiempo.com/colombia/cali/mas-mujeres-en-cali-se-mueven-en-moto-por-las-calles-de-cali-90766>.
- Ortiz Escalante, S., Ciocchetto, A., Fonseca, M., Casanovas, R., Valdivia, B., 2021. Movilidad cotidiana con perspectiva de género. Guía metodológica. CAF, Caracas. URL: <https://scioteca.caf.com/handle/123456789/1725>.
- Pérez, G., 2019. Políticas de movilidad y consideraciones de género en América Latina. Technical Report. CEPAL. URL: <https://repositorio.cepal.org/server/api/core/bitstreams/45670ca3-9bfa-48d6-b3e0-ebef7ffda908/content>.
- Pirra, M., Kalakou, S., Carboni, A., Costa, M., Diana, M., Lynce, A.R., 2021. A preliminary analysis on gender aspects in transport systems and mobility services: Presentation of a survey design. *Sustainability (Switzerland)* 13, 1–20. doi:[10.3390/su13052676](https://doi.org/10.3390/su13052676).
- Registro Único Nacional de Tránsito, 2025. Boletín de prensa 001 de 2025. balance del sector tránsito y transporte. URL: <https://www.runt.gov.co/sites/default/files/Bolet%C3%ADn%20de%20Prensa%20001%20de%202025.pdf>.
- Revista Motor, 2023a. Precios. URL: <https://www.motor.com.co/seccion/precios>.
- Revista Motor, 2023b. Soat 2023: así quedaron los precios para motos y carros. URL: https://www.motor.com.co/seccion/industria/soat-2023-asi-quedaron-los-precios-para-motos-y-carros_14914.
- Rosenbloom, S., 2006. Understanding women's and men's travel patterns: The research challenge, in: *Research on Women's Issues in Transportation*, Report of a Conference, Volume 1: Conference Overview and Plenary Papers, pp. 7–28. URL: <https://api.semanticscholar.org/CorpusID:130635294>.
- Rosenbloom, S., Burns, E., 1993. Gender differences in commuter travel in tucson: Implications for travel demand management programs. *Transportation Research Record* 1404, 82–90.
- S., M., Chalumuri (Ch.), R.S., 2016. Commuter's sensitivity in mode choice: An empirical study of New Delhi. *Journal of Transport Geography* 57, 207–217. URL: <https://www.sciencedirect.com/science/article/pii/S0966692316300199>, doi:<https://doi.org/10.1016/j.jtrangeo.2016.11.001>.
- Salazar-Serna, K., Barona, S.A., García, I.C., Cadavid, L., Franco, C.J., 2025. Addressing overfitting in classification models for transport mode choice prediction: a practical application in the aburrá valley, colombia. *Transportation Letters* 17, 1213–1230. URL: <https://doi.org/10.1080/19427867.2024.2422717>, doi:[10.1080/19427867.2024.2422717](https://doi.org/10.1080/19427867.2024.2422717), arXiv:<https://arxiv.org/abs/https://doi.org/10.1080/19427867.2024.2422717>.
- Salazar-Serna, K., Cadavid, L., Franco, C.J., Carley, K.M., 2023. Simulating transport mode choices in developing countries, in: Thomson, R., Al-khateeb, S., Burger, A., Park, P., A. Pyke, A. (Eds.), *Social, Cultural, and Behavioral Modeling*, Springer Nature Switzerland, Cham. pp. 209–218.

- Scheiner, J., Holz-Rau, C., 2012. Gendered travel mode choice: a focus on car deficient households. *Journal of Transport Geography* 24, 250–261. doi:10.1016/j.jtrangeo.2012.02.011.
- Scheiner, J., Holz-Rau, C., 2017. Women's complex daily lives: a gendered look at trip chaining and activity pattern entropy in germany. *Transportation* 44, 117–138. doi:10.1007/s11116-015-9627-9.
- Secretaría de Movilidad, 2024. Operativos de control vial: la apuesta de la alcaldía de cali para disminuir los altos índices de accidentalidad. <https://www.cali.gov.co/movilidad/publicaciones/182713/operativos-de-control-vial-la-apuesta-de-la-alcaldia-de-cali-para-disminuir-los-altos-indices-de-accidentalidad/>.
- Sharaby, N., Shiftan, Y., 2012. The impact of fare integration on travel behavior and transit ridership. *Transport Policy* 21, 63–70. doi:10.1016/j.tranpol.2012.01.015.
- Srinivasan, S., 2005. Influence of residential location on travel behavior of women in chennai, india, in: *Research on women's issues in transportation: report of a conference, November 18-20, 2004, Chicago, Illinois. Vol. 2 Technical papers*, pp. 4–13. URL: <https://api.semanticscholar.org/CorpusID:131665109>.
- Steiner, R., 1998. Women's travel for shopping in traditional neighborhoods: How does a woman's role in the household affect activity and travel for shopping?, in: *Women's Travel Issues Second National Conference*, pp. 299–313. URL: <https://www.fhwa.dot.gov/ohim/womens/chap16.pdf>.
- SURA, 2024. Seguros. URL: https://www.sura.co/seguros/personas/movilidad/motos/digital?utm_source=segurossura_personas&utm_medium=boton_home&utm_campaign=seguro_moto.
- Taylor, B.D., Ralph, K., Smart, M., 2015. What explains the gender gap in schlepping? testing various explanations for gender differences in household-serving travel. *Social Science Quarterly* 96, 1493–1510. doi:10.1111/ssqu.12203.
- Tilley, S., Houston, D., 2016. The gender turnaround: Young women now travelling more than young men. *Journal of Transport Geography* 54, 349–358. doi:10.1016/j.jtrangeo.2016.06.022.
- TomTom, 2025. Traffic index. URL: <https://www.tomtom.com/traffic-index/ranking/>.
- Train, K.E., 2009. *Discrete Choice Methods with Simulation*. 2nd ed., Cambridge University Press.
- Tu Carro, 2023. Motos. URL: <https://motos.tucarro.com.co/>.
- Turbo, R., 2023. ¿cada cuánto debería cambiar su carro? URL: <https://www.revistaturbo.com/cada-cuanto-deberia-cambiar-su-carro-1655/>.
- Urrea Giraldo, F., Candelo Álvarez, A.F., 2017. Cali, ciudad región ampliada: una aproximación desde la dimensión étnica-racial y los flujos poblacionales. *Sociedad y Economía* doi:10.25100/sye.v0i33.5628.
- Vance, C., Iovanna, R., 2007. Gender and the automobile: An analysis of non-work service trips. *Transportation Research Record: Journal of the Transportation Research Board* 2013, 54–61. URL: <https://doi.org/10.3141/2013-08>.
- Vanderschuren, M.J., Phayane, S.R., Gwynne-Evans, A.J., 2019. Perceptions of gender, mobility, and personal safety: South Africa moving forward. *Transportation Research Record* 2673, 616–627. doi:10.1177/0361198119854087.
- Wen, C.H., Wang, W.C., Fu, C., 2012. Latent class nested logit model for analyzing high-speed rail access mode choice. *Transportation Research Part E: Logistics and Transportation Review* 48, 545–554. URL: <https://www.sciencedirect.com/science/article/pii/S136655451100113X>, doi:https://doi.org/10.1016/j.tre.2011.09.002.
- Zeigermann, L., 2024. Mixlelast: Stata module for mixed logit elasticities and marginal effects. URL: <http://fmwww.bc.edu/repec/bocode/m/mixlelast.pdf>.
- Zhang, Y., Song, Y., 2024. Gender differences in travel and everyday life: A data-driven approach to address the intersectional nature of gender as a social construct. *Travel Behaviour and Society* 36. doi:10.1016/j.tbs.2024.100797.

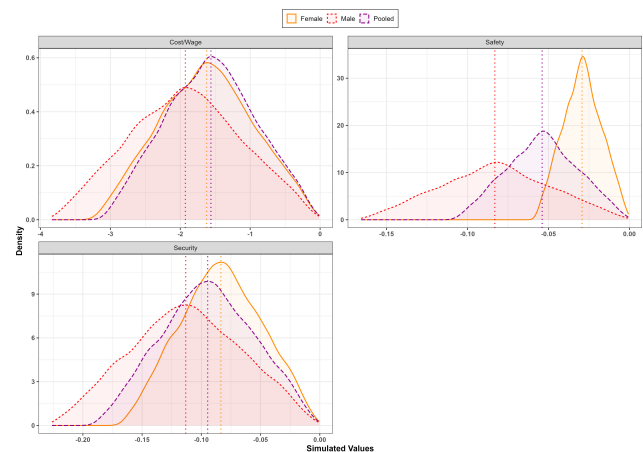


Figure 5: Triangular distribution: Total Travel Cost per Trip/Daily Wage, Safety, and Security by Model